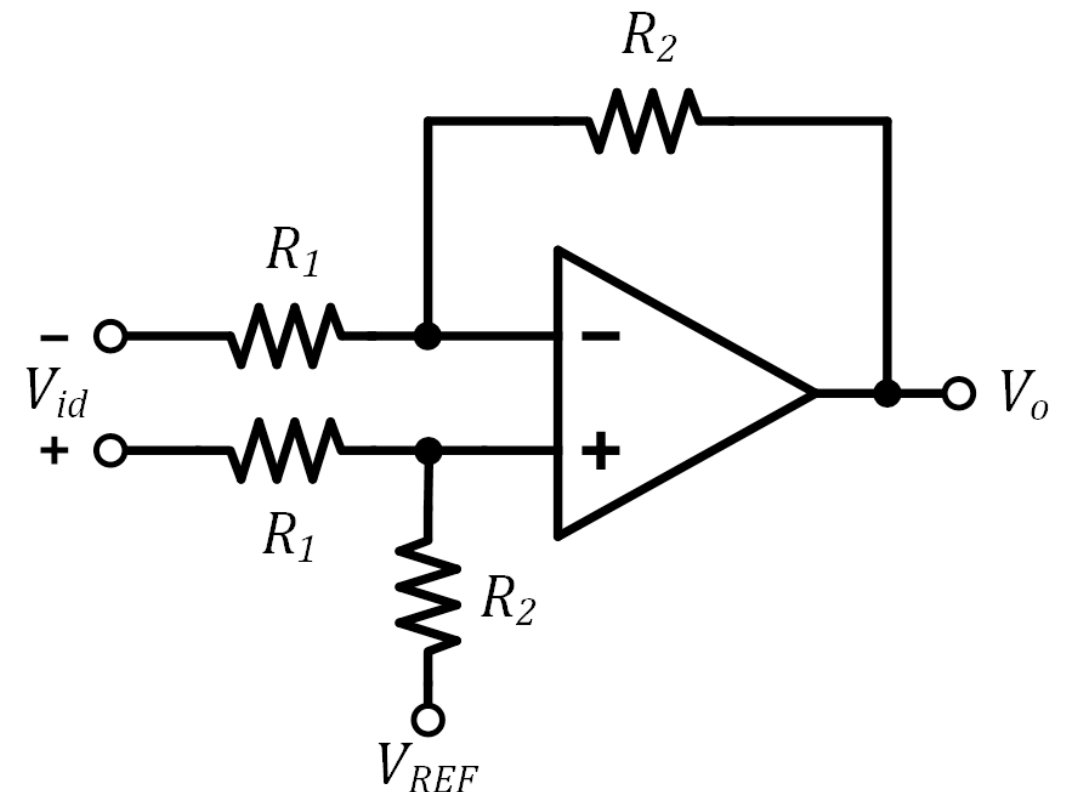


Difference Amplifier Reference Voltage

- For **single-ended power supply applications**, the **DC output** common-mode level is **set by V_{REF}**
- If V_{REF} is equal to the DC input common-mode, in the absence of an AC signal the current through R_1 and R_2 is zero and the output voltage is also V_{REF}
- If the current required by V_{REF} is small, it can be set by a voltage divider
- However, many applications may require a more stable voltage, necessitating a low-impedance voltage for V_{REF} (e.g. an additional opamp)



$$V_o = V_{REF} + \frac{R_2}{R_1} V_{id}$$

